General Relativity II / Cosmology

Fall 2021

Homework Assignment No. 2

Problems in Carroll's book:

Chapter 8

Problems 1, 2, 4, 5, 6, 7

Ex. 1 The expansion rate as a function of redshift depends upon the contents of the Universe as given by Eqn. (8.119) in the textbook.

(a) Find an analytical expression for $H_0 t_0$ for a Universe with matter and non-zero spatial curvature and no cosmological constant.

(b) Find an analytical expression for $H_0 t_0$ for a Universe with matter and non-zero cosmological constant and zero spatial curvature.

Hint: the solution to the integrals will involve either a logarithm or an inverse hyperbolic function.

(c) In each case above, what are the limits when $\Omega_m \to 0$ and when $\Omega_m \to 1$?